

Artificial Intelligence-Assisted Directional Prediction of Thermoelastic Wave Propagation in Geological Media

A comparative Beamer presentation for diploma defence

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Relevance of the Topic

Why this topic matters

- ▶ Thermally affected rocks are relevant to geothermal systems, underground engineering, and geophysical interpretation.
- ▶ Temperature change may alter stress, deformation, and wave behavior inside the geological medium.
- ▶ Repeated coupled simulations are informative, but expensive for comparative directional studies.

Working position of this thesis

This thesis is positioned as a *research prototype* for *artificial intelligence (AI)-assisted directional prediction* of thermoelastic wave propagation. It compares neural surrogate approaches under simplified assumptions rather than replacing numerical solvers.

Research Problem

Problem statement

The research problem is to estimate directional characteristics of thermoelastic wave propagation in geological media using an artificial intelligence (AI)-assisted comparative framework under simplified physical assumptions.

- ▶ The problem is important because coupled thermal and mechanical calculations can be costly when many directions or materials are compared.
- ▶ This work estimates selected source–probe responses and travel-time characteristics in a controlled model-based calculation setting.
- ▶ It does not claim a complete field-scale thermoelastic simulator or field-validated prediction system.

Aim and Tasks

Aim

To develop and justify an artificial intelligence (AI)-assisted framework for directional prediction of thermoelastic wave propagation in geological media.

1. Establish the physical and geological basis of the problem.
2. Formulate a simplified thermoelastic prediction task in two dimensions (2D).
3. Build a COMSOL-derived synthetic dataset for controlled comparison.
4. Compare Physics-Informed Neural Network (PINN), Fourier Neural Operator (FNO), MeshGraphNet, and Transformer-style surrogates.
5. Interpret the results, limitations, and next development steps.

The calculations were performed for representative geological materials, including granite, basalt, sandstone, and limestone.

Object and Subject of the Study

Object

Thermoelastic wave propagation in geological media represented by effective rock parameters.

Subject

artificial intelligence (AI)-assisted directional prediction of thermal and displacement response along a selected source–probe path.

Scope

The work does not claim to solve the full thermoelastic problem in real geological formations. It studies a simplified two-dimensional (2D) *comparative framework*.

Thermoelastic Coupling Logic

1. **Thermal excitation:** a local source changes the temperature field.
2. **Thermal expansion:** temperature change produces local strain.
3. **Stress formation:** constrained expansion creates thermal stress.
4. **Mechanical response:** stress gradients drive displacement.

$$T \longrightarrow \varepsilon_{\text{th}} \longrightarrow \sigma \longrightarrow u$$

Physical meaning

Temperature change causes thermal strain; thermal strain contributes to stress; stress affects displacement and wave propagation.

T – temperature [K]; ε_{th} – thermal strain [dimensionless]; σ – stress [Pa]; u – displacement [m].

Geological Media and Effective Parameters

Rock	ρ kg/m ³	V_P m/s	k W/(m K)	C_p J/(kg K)
Granite	2650	5850	2.50	850
Basalt	3000	6550	1.95	1300
Sandstone	2725	5000	2.25	1000
Limestone	2675	5400	2.55	1050

Effective scalar parameters support comparison under a locally homogeneous isotropic approximation, not field calibration.



Public-domain limestone texture, Wikimedia Commons / Titus Tschardtke.

Mechanical Rock Parameters

Material	Type	ρ kg/m ³	E GPa	ν –	μ GPa	K GPa	V_P/V_S m/s
Granite	intrusive	2650	76.28	0.245	30.63	49.84	5850/3400
Granodiorite	intrusive	2795	85.91	0.255	34.24	58.35	6100/3500
Basalt	volcanic	3000	103.96	0.266	41.07	73.95	6550/3700
Diabase	hypabyssal	2975	98.22	0.274	38.56	72.36	6450/3600
Gabbro	mafic	3075	114.27	0.287	44.40	89.33	6950/3800
Sandstone	sedimentary	2725	55.75	0.259	22.13	38.61	5000/2850
Limestone	carbonate	2675	61.48	0.277	24.08	45.90	5400/3000
Marble	metamorphic	2725	82.41	0.270	32.43	59.82	6150/3450
Schist	foliated	2800	68.20	0.267	26.91	48.82	5500/3100
Quartzite	quartz-rich	2650	83.50	0.250	33.40	55.70	6150/3550

Elastic constants are effective midpoint values derived under isotropic assumptions for model-ready comparison.

Thermal and Porosity Parameters

Material	α $10^{-6}/\text{K}$	k $\text{W}/(\text{m K})$	C_p $\text{J}/(\text{kg K})$	C $\text{MJ}/(\text{m}^3 \text{ K})$	γ MPa/K	Porosity %
Granite	7.9	2.50	850	2.25	1.18	total 45.5
Granodiorite	—	2.40	—	—	—	—
Basalt	—	1.95	1300	3.90	—	total 19.0
Diabase	—	2.50	850	2.53	—	—
Gabbro	—	2.30	1000	3.08	—	total 43.5
Sandstone	11.6	2.25	1000	2.73	1.34	total 31.5; eff. 26.5
Limestone	8.0	2.55	1050	2.81	1.10	total 31.5; eff. 18.0
Marble	9.8	2.80	850	2.32	1.76	—
Schist	—	2.75	1150	3.22	—	total 26.5; eff. 27.5
Quartzite	—	5.10	1000	2.65	—	—

Missing thermal expansion or coupling values are shown explicitly as dashes rather than inferred.

Rock Properties Controlling Propagation

Mechanical controls

- ▶ Density controls inertia.
- ▶ Elastic stiffness controls resistance to deformation.
- ▶ Porosity, cracks, and saturation change effective stiffness.

Thermal controls

- ▶ Conductivity controls heat transfer.
- ▶ Heat capacity controls thermal storage.
- ▶ Thermal expansion converts temperature change into strain.

Directional behavior can appear because bedding, foliation, aligned pores, fractures, and stress state make real rocks structurally ordered.

Elastic Waves in Geological Media

Compressional (P) waves

Compressional waves. Particle motion is mainly parallel to the propagation direction and involves volume change.

Shear (S) waves

Shear waves. Particle motion is mainly transverse to the direction of propagation and depends on shear rigidity.

$$V_P = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}, \quad V_S = \sqrt{\frac{G}{\rho}}$$

V_P, V_S – compressional and shear wave velocities [m/s]; K – bulk modulus [Pa]; G – shear modulus [Pa]; ρ – density [kg/m³]. These relations show that stiffness increases wave velocity, while density increases inertia.

Heat Conduction and Thermal Expansion

Fourier heat flux and heat equation

$$q = -k\nabla T, \quad \rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + Q$$

q – heat flux [W/m²]; k – thermal conductivity [W/(m K)]; T – temperature [K]; ρ – density [kg/m³]; C_p – specific heat capacity [J/(kg K)]; t – time [s]; Q – volumetric heat source [W/m³].

Temperature perturbation and thermal strain

$$\theta = T - T_0, \quad \varepsilon_{\text{th}} = \alpha \Delta T$$

$\theta, \Delta T$ – temperature change [K]; T_0 – reference temperature [K]; ε_{th} – thermal strain [dimensionless]; α – linear thermal expansion [1/K].

Key Physical Parameters

Thermal diffusivity relation

$$\alpha_T = \frac{k}{\rho C_p}$$

- ▶ α_T is thermal diffusivity [m^2/s].
- ▶ k is thermal conductivity [$\text{W}/(\text{m K})$].
- ▶ ρ is density [kg/m^3], and C_p is heat capacity [$\text{J}/(\text{kg K})$].
- ▶ Larger α_T means faster spreading of a temperature disturbance.

Thermoelastic Coupling

Thermoelastic extension of Hooke's law

$$\theta = T - T_0$$
$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + 2\mu \varepsilon_{ij} - \gamma \delta_{ij} \theta, \quad \gamma = 3K\alpha$$

Equation of motion

$$\rho \frac{\partial^2 u_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial x_j} + f_i$$

θ – temperature perturbation [K]; T, T_0 – current and reference temperatures [K]; σ_{ij} – stress tensor [Pa]; ε_{ij} – strain tensor [dimensionless]; ε_{kk} – volumetric strain [dimensionless]; λ, μ, K – Lamé and bulk moduli [Pa]; δ_{ij} – Kronecker delta [dimensionless]; α – thermal expansion [1/K]; γ – thermoelastic coupling coefficient [Pa/K]; u_i – displacement [m]; x_j – coordinate [m]; f_i – body force density [N/m³].

The relation states that stress depends on elastic deformation and temperature-induced expansion; this is why thermal and elastic fields are coupled in the model.

Mathematical Scope and Primary Fields

Simplified model scope

- ▶ 2D continuum: $\Omega \subset \mathbb{R}^2$, $(x, t) \in \Omega \times [0, T_f]$.
- ▶ In-plane displacement: $u(x, t) = [u, v]^T$.
- ▶ Small deformation and linear isotropic thermoelasticity.
- ▶ Pores, fractures, fluids, plasticity, damage, and full anisotropy are not resolved explicitly.

These assumptions keep the task suitable for comparative AI-assisted directional prediction and make the results interpretable within the scope of a bachelor thesis. Here Ω is the model domain [m²], x is position [m], t is time [s], T_f is final time [s], and u, v are displacement components [m].

Field	Meaning	Role
$T(x, t)$	Temperature	Thermal state
$\theta(x, t)$	Perturbation	Thermal loading
(u, v)	Displacement	Probe response
ε_{ij}	Strain	Deformation
σ_{ij}	Stress	Internal force
$d, \hat{d}, \phi$	Direction	Source–probe query

These fields connect the physical formulation with the prediction methodology used in the comparative prototype.

Directional Prediction Task



A source point and a probe point define the directional query.

$$d = x_p - x_s, \quad L = \|d\|, \quad \hat{d} = \frac{d}{\|d\|}$$

$$\phi = \text{atan2}(d_y, d_x), \quad \mathcal{F} : X \rightarrow \hat{Y}$$

x_s, x_p – source and probe positions [m]; d – direction vector [m]; L – distance [m]; $\hat{d}$ – unit direction [dimensionless]; ϕ – azimuth [rad]; $\mathcal{F}$ maps input features X to predicted outputs $\hat{Y}$.

Prediction contract

Input: material parameters, geometry, direction, and time. Output: estimated temperature, displacement, directional response, and travel time.

COMSOL-Based Synthetic Dataset

Controlled thermoelastic calculation

Two-dimensional (2D) plane-strain rock domain $1\text{ m} \times 1\text{ m}$ with a heated internal source used to generate time-dependent synthetic data.

- ▶ Initial rock temperature: 293.15 K
- ▶ Source temperature: 1500 K
- ▶ Outputs exported per node and time step: T , (u, v) , velocity, stress, strain, and material features
- ▶ Purpose: create a common reference dataset for model comparison

Data Representation and Preprocessing

- ▶ COMSOL comma-separated value (CSV) exports are aligned into a single node set.
- ▶ Dynamic features are assembled into a spatio-temporal tensor:

$$\mathbf{S} \in \mathbb{R}^{N_t \times N_p \times F_d}$$

- ▶ Static material and geometry features are stored separately.
- ▶ Feature channels are normalized before training:

$$\tilde{x} = \frac{x - \mu}{\sigma + \varepsilon}$$

- ▶ The same processed dataset supports coordinate, grid, and graph models.

$\mathbf{S}$ is the assembled data tensor; N_t is the number of time steps, N_p the number of spatial points, and F_d the number of dynamic feature channels. $\tilde{x}$ is a normalized feature [dimensionless], x is the original feature in its native unit, μ and σ are its dataset mean and standard deviation, and ε is a small dimensionless stabilizer.

Comparative Prototype Framework

One workflow for all model routes

The frontend, backend orchestrator, material catalog, and model services use one shared input–output contract for fair comparison.

- ▶ Same source–probe geometry for all routes
- ▶ Same effective geological material presets
- ▶ Same normalized response format: temperature, displacement, direction, travel time, and diagnostics

Shared Model Comparison Contract

Common interpretation

All routes are treated as surrogate modelling approaches for the same AI-assisted directional prediction task, not as complete numerical solvers.

Input

Material parameters, source–probe geometry, observation time, and processed field features.

Output

Estimated temperature, displacement components, travel time, and diagnostic response quantities.

Comparison basis

Same material presets, geometry contract, response format, and analytical travel-time sanity check.

Physics-Informed Neural Network (PINN)

Role

Physics-informed baseline in the comparison.

Input / output

Input: coordinates, time, and material parameters. Output: estimated temperature and in-plane displacement.

Why and limitation

Used because thermoelastic residuals can be included in training. It is still a baseline model, not an exact equation solver.

Model relation

$$f_{\theta}(x, y, t, E, \nu, \rho, \alpha, k, C_p) \rightarrow (T, u, v)$$

$$\mathcal{L} = \lambda_{\text{sup}} \mathcal{L}_{\text{sup}} + \lambda_{\text{vel}} \mathcal{L}_{\text{vel}} + \lambda_{\text{el}} \mathcal{L}_{\text{elastic}} + \lambda_{\text{th}} \mathcal{L}_{\text{thermal}}$$

x, y – coordinates [m]; t – time [s]; E – Young's modulus [Pa]; ν – Poisson's ratio [dimensionless]; ρ – density [kg/m^3]; α – thermal expansion [$1/\text{K}$]; k – conductivity [$\text{W}/(\text{m K})$]; C_p – heat capacity [$\text{J}/(\text{kg K})$]; T, u, v – predicted temperature [K] and displacements [m]. $\mathcal{L}$ and $\mathcal{L}_{\cdot}$ are normalized loss terms; $\lambda_{\cdot}$ are dimensionless loss weights.

Fourier Neural Operator (FNO)

Role

Neural operator surrogate for structured two-dimensional fields.

Input

Grid-based material and response features, together with the shared source–probe and time query context.

Output

Estimated field-style response used to derive directional quantities.

Why and limitation

Included because Fourier operators can capture broad field patterns. In the current prototype, its output scale needs recalibration.

MeshGraphNet

Role

Graph-based surrogate model for mesh-aware prediction.

Input

Mesh nodes, graph connectivity, material features, and query context.

Output

Node-level or probe-level response estimates after graph message passing.

Why and limitation

Included because graph structure can represent local mesh interactions. The current route behaves mainly as a fixed-horizon predictor.

Transformer-Style Surrogate Model

Role

Attention-based baseline surrogate for fast query prediction.

Input

Tokens describing geometry, material parameters, state variables, and requested observation time.

Output

Fast estimates of temperature, displacement, travel time, and diagnostics.

Why and limitation

Included as a flexible low-latency route. Its physics priors are limited, so results require cautious interpretation.

Calculation Status

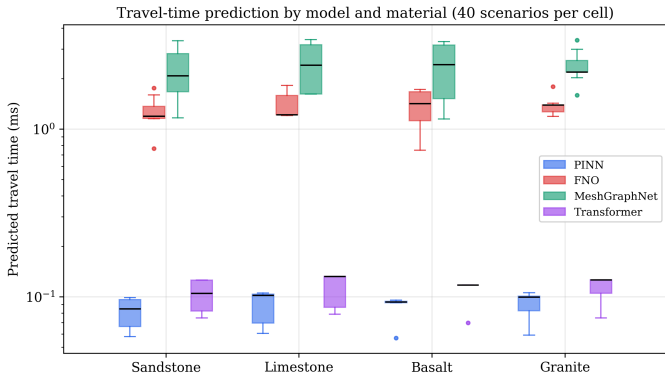
Calculation setup

1. **40 scenarios per material**, four materials: granite, sandstone, limestone, and basalt.
2. $40 \times 4 \times 4 = 640$ **model responses** evaluated under one shared contract.
3. **Zero fallback**: every plotted response is an active model output, not a mock.
4. **2D plane-strain only**: the operating envelope of all trained surrogates.

FNO caveat

FNO is included in the comparison, but its output scales are still miscalibrated. It should currently be treated as a diagnostic branch.

Travel-Time Predictions

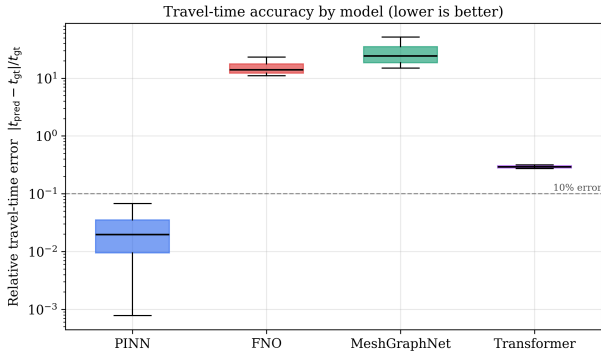


Reading the plot

- ▶ PINN ≈ 0.11 ms
- ▶ Transformer ≈ 0.14 ms
- ▶ FNO ≈ 1.29 ms
- ▶ MeshGraphNet ≈ 2.47 ms

Model-family differences are larger than the basalt–sandstone material difference in the current prototype.

Accuracy Against Analytical Reference



Analytical reference

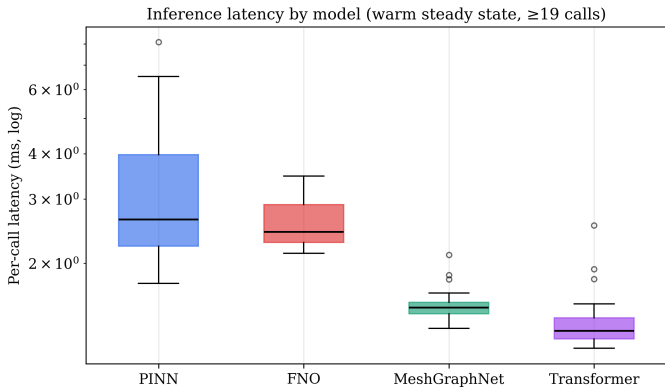
$$t_{\text{gt}} = \frac{r}{V_p}$$

t_{gt} – reference travel time [s]; r – source–probe distance [m]; V_p – compressional wave velocity [m/s].

Model	Median error	$\leq 10\%$
PINN	2.0%	100%
Transformer	29.2%	0%
FNO	1403.6%	0%
MeshGraphNet	> 100%	0%

PINN is the only route that remains inside the 10% band across the balanced sweep.

Inference Latency and Time Behavior



Median latency

- ▶ Transformer ≈ 1.3 ms
- ▶ MeshGraphNet ≈ 1.5 ms
- ▶ FNO ≈ 2.4 ms
- ▶ PINN ≈ 2.6 ms

PINN is also the only route that changes smoothly with requested observation time; the other routes currently behave as fixed-horizon predictors.

Threats to Validity

1. **Limited training data.** The COMSOL-derived pilot configuration is not a full calibrated geological database.
2. **Simplified 2D assumptions.** The study uses effective material parameters in a plane-strain setup.
3. **FNO normalization issue.** The current 2D adaptation is not yet physically reliable.
4. **Analytical reference is isotropic.** r/V_p is a useful sanity check, not a complete thermoelastic validation.

Conclusions

1. The thesis develops a physically grounded *comparative framework* for AI-assisted directional prediction.
2. PINN is the strongest *physics-informed baseline* in the current comparison and shows the best agreement with the analytical travel-time reference.
3. Transformer is the fastest route, while MeshGraphNet remains a useful mesh-aware surrogate candidate.
4. FNO should currently be treated as a diagnostic branch until output normalization is corrected.

Future Work

1. Retrain and recalibrate the 2D FNO route.
2. Add time-conditioned rollout behavior to MeshGraphNet.
3. Extend the material catalog with more complete thermoelastic parameters and validation data.
4. Compare against additional analytical or numerical baselines.

Final message

The prototype estimates directional characteristics of thermoelastic wave propagation within the simplified assumptions of the study.

Thank You

Questions?